

Design FMEA — T4R1FLASH (qualitative version, based on simulation signals)

Case: `t4\_r1\_flash` | Generation date: 2026-08-25 | Doc: VBF-T4R1FLASH-DFMEA-01

Qualitative caliber: S/O rated High/Medium/Low (H=3, M=2, L=1); RPN = S x O, simplified qualitative matrix.

Failure mode	Failure cause	Simulation signal (detectability basis)	Severity	Occurrence	RPN	Design-side mitigation (implemented)

t limitation / anode overpotential < 0 V at 4C		anode_potential_v min = +0.0193 V (baseline was -0.44 V)	H	L (post-mitigation)	3	Electrolyte transport override sign
Charge heat accumulation exceeding cooling		T_max 329.61 K vs 333.15 K limit (baseline 350.6 K)	H	L (post-mitigation)	3	Forced-air cooling h=40 W/m2K; overc
age beyond electrolyte stability window		HOMO/IE-EA vs 4.2 V window (molecular funnel; FEC/VC screened, 2 passed)	M	L	2	Additive screening at Stage 2; final
sufficient capacity	Electrode loading / N-P mismatch	1C capacity 3.955 Ah vs nominal 3.93 Ah	M	L	2	Thickness re-balance 60/68 µm, C_nom reset to true 1C (F

Conclusion

Highest-risk items: plating and thermal runaway — both mitigated in the final design with simulation-verified margins (see DVPR).
Complete FMEA including process/supplier failures: N/A (beyond pure simulation boundary).